*CIS BUSINESS ECONOMICS - WEEK 1: Energy Transmission Channel & Imported Inflation*

# THE ENERGY TRANSMISSION CHANNEL & IMPORTED INFLATION

*Tracing a USD 10/Barrel Brent Crude Shock Through India's WPI, CPI and Sectoral Cost Base*

CIS Business Economics (BE-2) - Week 1 Policy / Economics Research Submission

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**Core Question**

***"How does a USD 10/barrel spike in Brent crude filter into India's domestic WPI and CPI inflation, and which domestic industries are most vulnerable?"***

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## List of Abbreviations

| Abbrev. | Full Form | Abbrev. | Full Form |
|---|---|---|---|
| **ATF** | Aviation Turbine Fuel | **MoF** | Ministry of Finance |
| **BoP** | Balance of Payments | **MOSPI** | Ministry of Statistics & Programme Implementation |
| **CAD** | Current Account Deficit | **OMC** | Oil Marketing Company |
| **COGS** | Cost of Goods Sold | **PED** | Price Elasticity of Demand |
| **CPI** | Consumer Price Index | **PMUY** | Pradhan Mantri Ujjwala Yojana |
| **DGCA** | Directorate General of Civil Aviation | **PPAC** | Petroleum Planning & Analysis Cell |
| **EIA** | U.S. Energy Information Administration | **RBI** | Reserve Bank of India |
| **ERPT** | Exchange Rate Pass-Through | **SPR** | Strategic Petroleum Reserve |
| **FDI** | Foreign Direct Investment | **USD** | United States Dollar |
| **FY** | Financial Year | **VAT** | Value-Added Tax |
| **GDP** | Gross Domestic Product | **WPI** | Wholesale Price Index |
| **IEA** | International Energy Agency | **bbl** | Barrel (of crude oil) |
| **MPC** | Monetary Policy Committee | **pp** | Percentage Points |

# 1. Executive Summary

Roughly one-fifth of global seaborne crude oil transits the Strait of Hormuz, and the large majority of India's Gulf-origin crude passes through it. This report models the transmission of a marginal **USD 10/barrel** increase in Brent crude - set within a hostilities-driven crisis band of **USD 110**-120/barrel - through the Indian economy, given that India imports **87.7%** of the crude oil it consumes. The analysis traces an eleven-stage chain running from the oil-market shock to the annual **import bill**, the **Current Account Deficit**, the exchange rate, Wholesale and Consumer Price Index inflation, **GDP growth**, and sector-level cost stress, and concludes with a set of sequenced, trigger-based policy recommendations for the **RBI**, the **Ministry of Finance** and allied institutions.

## Key Findings - Base Case Impact of a +USD 10/bbl Shock

| Metric | Estimated Impact | Direction |
|---|---|---|
| **Annual oil import bill** | +USD 17.1 billion | Increase |
| **Current Account Deficit** | +0.44 percentage points of GDP | Widens |
| **USD/INR exchange rate** | +2.2% (Rs. 86.0 - > Rs. 87.9/USD) | Depreciation |
| **WPI inflation** | +0.56 to +0.65 percentage points | Increase |
| **CPI inflation (12-month)** | +0.45 to +0.50 percentage points | Increase |
| **Real GDP growth** | -0.15 to -0.25 percentage points | Drag |
| **Fiscal deficit** | +0.13 percentage points of GDP | Widens |

*Table 1. Base Case summary of the modelled transmission results. See Sections 5-7 for the full quantitative build-up and Section 14 for the calculation basis of every figure.*

This transmission is uneven across the economy. Aviation, petrochemicals and road logistics absorb the sharpest direct cost increases - between **2.4%** and **4.8%** of their operating cost base - because of high crude-linked input shares combined with weak pricing power, whereas power and steel are comparatively insulated by their reliance on coal. India's macro-buffers, including **forex reserves** in the **USD 655**-660 billion range, provide a genuine but time-limited cushion. The critical policy variable is **shock duration rather than shock magnitude**: within six weeks the shock is manageable using existing buffers (Best Case); **beyond 90 days, the growth-inflation trade-off intensifies** and the RBI's rate-cut cycle would likely need to pause to defend both price stability and currency credibility (Worst Case).

The report is structured to let a time-constrained evaluator verify the central claim quickly: Section 5 establishes the causal transmission chain, Sections 6-7 build up the quantitative results stage by stage, Section 8 translates the macro shock into sector-level cost stress, Section 9 stress-tests the Base Case against Best and Worst Case alternatives, and Section 10 converts the analysis into concrete, trigger-based policy actions for named institutions.

# 2. CIS Question, Scope & Methodology

**CIS Question:** How does a USD 10/barrel spike in Brent crude filter into India's domestic Wholesale Price Index (WPI) and Consumer Price Index (CPI) inflation, and which domestic industries are most vulnerable?

This report answers that question through a first-principles build-up: volumes, prices and elasticities are combined stage by stage rather than asserted top-down. Three scenarios - Best, Base and Worst Case - are modelled with probability weights of **20%**, **55%** and **25%** respectively, reflecting the assignment's premise of active Strait of Hormuz hostilities rather than a routine price fluctuation. The scope is deliberately confined to transmission mechanics and sectoral impact; the fiscal and external-financing consequences are developed further in the Week 2 report (Twin Deficits), and the capital-flow and reserve-adequacy consequences in the Week 3 report (Financial Contagion).

### 2.1 Analytical Approach

The quantitative method follows a consistent chain for every major result in this report: a claim is stated, the supporting evidence is identified and sourced, the governing economic mechanism is named, the calculation is shown explicitly with its formula and inputs, the result is interpreted in plain terms, and - where relevant - the policy implication is drawn out. Every material input is traceable to one of four categories, disclosed transparently through source attribution and methodology notes rather than through embedded classification tags: official data reported by RBI, PPAC, MOSPI and allied institutions; historical precedents used as benchmarks; values the report derives itself through explicit formulas; and scenario or sensitivity assumptions introduced because no real-time, shock-specific Indian estimate exists. Section 14 consolidates the full evidentiary basis for every material figure so that each claim in the main text can be independently verified.

### 2.2 Confidence and Limitations

Direction of effect - the CAD widens, WPI and CPI rise, **GDP growth** slows - is held with high confidence, since it follows directly from India's structural oil-import dependency. The exact magnitude of each effect is held with moderate confidence, since it depends on calibrated elasticities (price pass-through, exchange-rate sensitivity) for which no real-time, shock-specific Indian estimate exists; these are disclosed as scenario inputs rather than presented as observed facts. Geopolitical **shock duration** and intensity cannot be forecast with precision, which is why this report uses a three-scenario range with explicit probability weights rather than a single point estimate. Where later reports in this CIS use a different reference date for a given variable, this is a legitimate consequence of each week's assignment premise rather than an inconsistency; Section 14.1 reconciles this explicitly.

## 3. India's Energy Exposure - Baseline & Context

### 3.1 Macroeconomic Dashboard: Pre-Shock Baseline

All figures below are dated to the reference period stated in the Data, Evidence & Calculation Notes (Section 14), consistent with PPAC's FY24 import-dependency release and the RBI's latest published Weekly Statistical Supplement and MPC resolution at the time of writing.

| Indicator | Value | Basis |
|---|---|---|
| **Brent crude (pre-shock reference)** | ~USD 78/bbl | Recent market reference |
| **Brent crude (crisis scenario band)** | USD 110-120/bbl | Assignment premise |
| **India crude oil import dependency** | 87.7% | PPAC, FY24 |

| Indicator | Value | Basis |
|---|---|---|
| **India crude oil imports (annual)** | 232.7 million tonnes (~1,705.7 million barrels) | PPAC |
| **India oil import bill (last full FY)** | ~USD 132.4 billion | PPAC / Ministry of Petroleum |
| **India nominal GDP** | ~USD 3.9 trillion | MOSPI / RBI |
| **Forex reserves** | ~USD 655-660 billion | RBI Weekly Statistical Supplement |
| **Strategic Petroleum Reserve** | ~9.5 million tonnes (~10 days cover) | Ministry of Petroleum |
| **Current Account Deficit (baseline)** | ~1.1% of GDP (~USD 43 billion) | RBI, latest reported quarter |
| **USD/INR spot** | ~Rs. 86.0/USD | RBI Reference Rate |
| **Repo rate** | 5.50% | RBI MPC, latest resolution |
| **CPI inflation (run-rate baseline)** | ~3.5% | MOSPI, latest monthly print |
| **WPI inflation (run-rate baseline)** | ~2.5% | Office of the Economic Adviser |
| **Fiscal deficit (target)** | 4.4% of GDP (FY26 BE) | Union Budget |
| **Petrol / diesel retail (Delhi)** | Rs. 94.72 / Rs. 87.62 per litre | OMC pricing |

*Table 2. Pre-shock macroeconomic baseline used throughout this report. Full source detail and reference-period reconciliation across the four CIS weekly reports is provided in Section 14.*

## 3.2 Long-Term Structural Context

The oil-intensity of Indian GDP has declined gradually over the past decade as the services sector's GDP share has grown and energy efficiency has improved, which partially cushions the economy against a shock of any given size relative to a decade ago. Two structural factors nonetheless keep exposure high. First, EV and alternative-fuel penetration remains low - roughly **2-3% of new vehicle sales** - so the transport sector's crude dependence will persist for years regardless of policy intent. Second, the **Strategic Petroleum Reserve**, while it exists, covers only around 10 days of net imports against the IEA's 90-day benchmark for member countries, leaving limited tactical buffer relative to global peers. These structural facts frame why this report treats accelerated diversification (Section 11) as the primary lever available to reduce exposure to the next shock, rather than treating the current shock in isolation.

## 3.3 Geopolitical Context and Oil Market Mechanics

The Strait of Hormuz is a 33-kilometre chokepoint carrying roughly **20%** of the world's daily petroleum liquids consumption, along with the large majority of the crude India sources from Saudi Arabia, Iraq, the UAE and Kuwait. A blockade or sustained hostilities would transmit into oil prices through three simultaneous channels: physical supply disruption - tanker delays, war-risk insurance, and the limited 1.5-1.8 million-barrels-per-day bypass capacity of the Fujairah pipeline against the 17-20 million barrels per day that normally transit the Strait; repricing of risk premia, historically adding **USD 5**-15/barrel during comparable episodes (the 1990-91 Gulf conflict, the 2019 tanker incidents, the 2024 Red Sea disruptions); and elevated freight and war-risk marine insurance costs that compound the **landed cost** independent of the FOB crude price.

The most policy-relevant question is not the absolute price level but the marginal propagation of each incremental **USD 10/barrel** move within the elevated crisis band. This report therefore isolates the transmission effect of a move from **USD 110** to **USD 120** per barrel.

**Figure 1 — Brent Crude Price Trajectory and Crisis Scenario Band**




***Figure 1.*** *Brent Crude Price Trajectory and Crisis Scenario Band.* ***Source/Method:*** *Author's construction based on the stated pre-shock reference and the assignment's crisis-band premise; illustrative post-shock path shown for context.* ***Interpretation:*** *The figure isolates the USD 110-120/bbl crisis band used for the marginal shock analysis; this report's calculations fix the marginal move at +USD 10/bbl within that band.*

## 4. Business Economics Framework

The transmission mechanism modelled in this report rests on a small number of core Business Economics concepts, each applied quantitatively rather than merely referenced. Table 3 summarises each concept alongside its specific application in this report and the section where it is developed in full.

| Concept | Application in This Report |
|---|---|
| **Supply shock / short-run inelasticity** | India is a price-taker with an 11-12% domestic crude self-sufficiency ratio; a geopolitical supply shock transmits almost fully into price rather than being absorbed through quantity or domestic substitution (Section 5.3, Stage 1). |
| **Price elasticity of demand (PED)** | A short-run crude PED of -0.05 quantifies the marginal demand-destruction offset to the import-bill increase (Section 5.3, Stages 2-3; Section 6.1). |
| **Cost-push / imported inflation** | The rise in landed crude cost raises producer input costs (WPI) before flowing into consumer prices (CPI) - the textbook cost-push channel (Section 7). |
| **Exchange-rate pass-through (ERPT)** | Rupee depreciation compounds the USD-denominated price shock, so the INR-terms landed-cost increase (+11.5%) exceeds the USD-terms increase (+9.1%) (Section 5.3, Stage 6). |
| **Aggregate supply / producer cost transmission** | WPI, capturing ex-factory and wholesale prices before retail taxation, shows higher elasticity to the shock than CPI, which embeds a large fixed-tax wedge (Section 7.1-7.2). |
| **Balance of Payments identity** | An uncompensated rise in the import bill directly widens the trade deficit and, absent offsetting inflows, the Current Account Deficit (Section 5.3, Stage 4). |
| **Inflation-growth trade-off** | Second-round inflation effects and higher input costs feed into the GDP growth drag (Section 6.4), foreshadowing the central Week 4 policy tension. |

*Table 3. Core Business Economics concepts applied quantitatively in this report, each traceable to the section that develops it.*

# 5. End-to-End Transmission Mechanism

The shock propagates through eleven distinct stages over roughly two to four quarters. This section establishes the causal chain that links them; full calculations for the quantitatively central stages (**import bill**, CAD, exchange rate, WPI, CPI) are developed in Sections 6-8.

## 5.1 Master Transmission Diagram

Middle East hostilities transmit into a global Brent price spike, which raises India's oil import bill (+**USD 17.1 billion**), widens the **Current Account Deficit** (**+0.44 pp** of GDP), and puts depreciation pressure on the Rupee (**+2.2%**). This compounds into **imported inflation** - a landed-cost increase of **+11.5%** in INR terms - which transmits into WPI (**+0.56 to +0.65 pp**) and CPI (**+0.45 to +0.50 pp**), producing a measurable **GDP growth** drag (-0.15 to -0.25 pp) that requires a coordinated RBI/Government policy response and, over the long run, a structural energy-diversification response.

**Figure 2 — Master Macroeconomic Transmission Diagram: Middle East Shock to Domestic Impact**

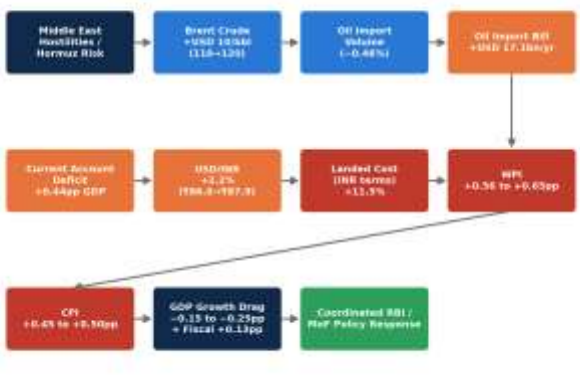



***Figure 2.*** *Master Macroeconomic Transmission Diagram: Middle East Shock to Domestic Impact.* ***Source/Method:*** *Author-constructed flow diagram synthesising Sections 5-8.* ***Interpretation:*** *The diagram traces the causal path from the external geopolitical trigger through six intermediate macro variables to the domestic inflation and growth outcomes; each stage is developed individually below.*

## 5.2 The Eleven-Stage Chain - Quick Reference

| Stage | Transition | Headline Result |
|---|---|---|
| **1** | Conflict - > global oil prices | Brent +USD 10/bbl within the crisis band |
| **2** | Global prices - > import volume response | Volume -0.46% (marginal, price-dominant) |
| **3** | Import volume/price - > oil import bill | +USD 17.1 billion/year |
| **4** | Import bill - > Current Account Deficit | +0.44 pp of GDP |
| **5** | CAD - > exchange rate | USD/INR +2.2% |
| **6** | Exchange rate + price - > landed cost | +11.5% in INR terms |
| **7** | Landed cost - > WPI | +0.56 to +0.65 pp |

| Stage | Transition | Headline Result |
|---|---|---|
| **8** | WPI/landed cost - > CPI | +0.45 to +0.50 pp |
| **9** | CPI - > corporate/retail cost pass-through | +4.2 to +4.3% retail petrol |
| **10** | Cost pass-through - > household purchasing power | 1.5-1.8× headline CPI for low-income households |
| **11** | Cumulative effect - > GDP and fiscal position | -0.15 to -0.25 pp GDP; +0.13 pp fiscal deficit |

*Table 4. The eleven-stage transmission chain, summarising the transition and headline quantitative result at each stage.*

**Figure 3 — The Eleven-Stage Transmission Chain**

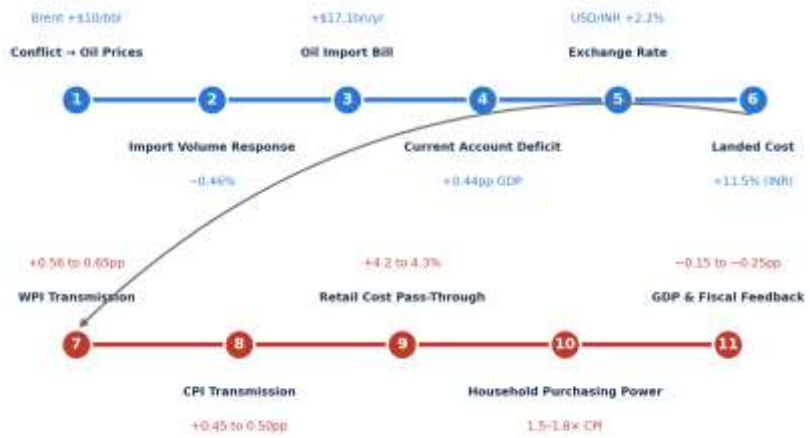



***Figure 3.*** *The Eleven-Stage Transmission Chain.* ***Source/Method:*** *Author's construction, sequencing the results in Table 4 as a linear chain.* ***Interpretation:*** *Stages 1-6 (blue) are the external/financial transmission leg; Stages 7-11 (red) are the domestic inflation, household and growth leg. The chain is sequential but each stage's magnitude is calculated independently in Sections 6-8.*

## 5.3 Stage-by-Stage Analysis

### Stage 1 - Middle East Conflict - > Global Oil Prices

Oil is priced at the margin by a supply base that is geographically concentrated and inelastic (OPEC+ and Gulf producers routing through Hormuz) meeting demand that is equally inelastic in the short run. Supply-side geopolitical shocks therefore transmit almost fully into price volatility rather than being absorbed by quantity adjustment. Brent moves from the ~**USD 78/bbl** reference to the stated crisis band of **USD 110**-120/bbl; this report's marginal analysis fixes the move at +**USD 10/bbl** within that band (110 - > 120). India cannot offset this through domestic supply - it produces ~0.6 million barrels/day against consumption of ~5.3 million barrels/day, an **11-12% self-sufficiency ratio**. The IEA and EIA monitor global inventories and OPEC+ spare capacity (~4-5 million barrels/day theoretical), though this capacity is itself largely Gulf-concentrated and can be trapped behind the same chokepoint; non-Gulf spare capacity (US shale, Brazil, Guyana) typically needs 6-12 months to respond. The principal mitigation available to India is participation in coordinated, IEA-aligned strategic reserve releases and continued diversification of non-Gulf crude sourcing.

### Stage 2 - Global Oil Prices - > Indian Oil Imports (Volume Response)

The short-run **price elasticity** of demand for crude is very low because transport fuel, LPG and petrochemical feedstock have few short-run substitutes. Using a short-run PED of -0.05: % change in Qd = PED × % change in P = -0.05 × (10/110) = -0.05 × **9.09%** = **-0.455%**. This implies a volume reduction of ~7.75 million barrels/year against the 1,705.7 million-barrel annual base - an adjusted volume of 1,697.95 million barrels/year. Demand destruction is real but marginal (under **0.5%** of volume); price, not volume, is the dominant transmission variable in the short run. PPAC monitors monthly import volumes, and Indian refiners retain some flexibility by shifting toward discounted grades (e.g., Russian Urals) to partially offset Gulf-linked price increases. The principal risk is a physical, not merely price-based, supply constraint from a genuine blockade, which would force reliance on a narrower non-Gulf supplier base; the recommended mitigation is long-term supply contracts and continued sourcing diversification (Section 11).

### Stage 3 - Oil Import Bill Calculation

Treating the import bill as price times volume, given the short-run inelasticity established in Stage 2, the unadjusted first-order calculation is Change in Bill = Change in P × Q = **USD 10** × 1,705.7 million barrels = **USD 17.06 billion**/year. An elasticity-adjusted, second-order calculation gives $Bill_1$ (at **USD 110/bbl**) = **USD 187.63 billion** and $Bill_2$ (at **USD 120/bbl**, adjusted volume) = **USD 203.75 billion**, for a net increase of USD 16.13 billion/year (+9.1% over the USD 110/bbl baseline bill). This report uses the unadjusted USD 17.1 billion figure as the headline Base Case number, consistent with - and the conservative upper bound of - the elasticity-adjusted range of USD 16.1-17.1 billion, since it does not assume any demand-destruction offset. PPAC and the **Ministry of Petroleum** publish revised import-bill projections as conditions change; the principal tail risk is a further overshoot toward USD 130-140/bbl, where compounding freight and war-risk insurance premiums would escalate the bill non-linearly beyond simple arithmetic projection.

**Figure 4 — Oil Import Bill Decomposition**




*__Figure 4.__ Oil Import Bill Decomposition. __Source/Method:__ Author's calculation per the formulas in Stage 2-3 above, applied to the PPAC baseline import volume. __Interpretation:__ The gross price effect (+USD 17.06bn) is partly offset by a small elasticity-driven demand-destruction effect, leaving a net elasticity-adjusted increase of USD 16.13bn - confirming that price, not volume, dominates the transmission.*

### Stage 4 - Current Account Deficit Impact

Applying the Balance of Payments identity, an uncompensated rise in the oil import bill directly widens the trade deficit and, absent an offsetting rise in exports, remittances or invisibles, the Current Account Deficit.

$CAD_2$(% GDP) = [$CAD_1$(USD) + Change in Import Bill] / GDP. Substituting $CAD_1$ = **1.10%** × **USD 3,900 billion** = **USD 42.9 billion** (~ **USD 43 billion**) and Change in Import Bill = **USD 17.06 billion**: $CAD_2$ = (43 + 17.06)/3,900 = **USD 60.06 billion** / USD 3,900 billion = 1.54% of GDP, i.e. Change in CAD = +0.44 percentage points of GDP. It is worth being explicit that the Balance of Payments identity establishes an accounting relationship, not a causal proof on its own: the identity tells us that a larger import bill mechanically requires a larger offsetting financing flow (reserve drawdown, capital inflow, or **currency depreciation**) somewhere in the accounts; it **does not by itself specify which of those three absorbs the adjustment** - that is an empirical, institutional question addressed in Stage 5. A widening from 1.1% to 1.5% of GDP remains within historically manageable territory - India's CAD peaked near 4.8% of GDP during the 2013 Taper Tantrum - but the direction and speed of deterioration are what drive currency volatility and sovereign-risk sentiment, not the level alone.

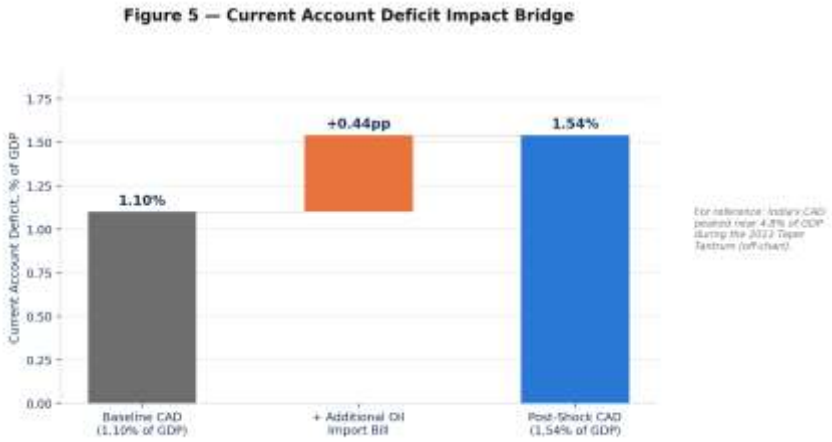



***Figure 5.*** *Current Account Deficit Impact Bridge.* ***Source/Method:*** *Author's calculation applying the Balance of Payments identity to the Stage 3 import-bill result.* ***Interpretation:*** *The additional oil import bill mechanically adds 0.44 percentage points to the CAD; the RBI's Department of Economic and Policy Research revises BoP projections on this basis, and the Ministry of Finance monitors the financing requirement against FDI/FPI inflows (developed further in the Week 3 report).*

### Stage 5 - Exchange Rate Impact

In India's managed-float regime, a widening CAD raises demand for foreign currency to settle the larger import bill, creating depreciation pressure that is partially offset by **RBI intervention** and capital inflows. Using the stylised sensitivity relationship % change in USD/INR = β × Change in CAD(% GDP), with a Base Case β of 5 (historical stress-episode range: 3-8): % change in USD/INR = 5 × **0.44%** = **2.2%**, giving a new rate of 86.0 × 1.022 = **Rs. 87.9/USD**. At β = 3 (Best Case), the rate moves to **Rs. 87.1/USD**; at β = 8 (Worst Case), to **Rs. 89.0/USD**. This β parameter is a stylised, single-factor calibration rather than an econometrically estimated pass-through coefficient, and should be read as an illustrative sensitivity range. The RBI's Financial Markets Operations Department intervenes via spot and forward operations to manage excessive volatility - not to defend a fixed level. India's **USD 655**-660 billion reserve buffer (each 1% of GDP of intervention capacity ~ USD 39 billion) provides ample tactical capacity for a shock of this magnitude.

**Figure 6 — USD/INR Sensitivity to the FX-Beta Parameter**

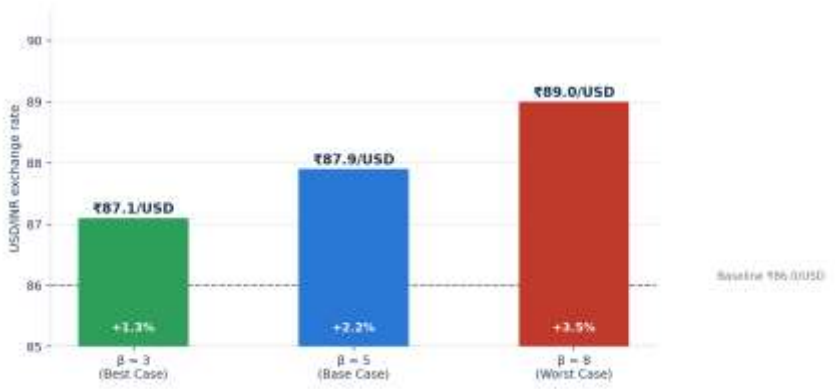



***Figure 6.*** *USD/INR Sensitivity to the FX-Beta Parameter.* ***Source/Method:*** *Author's calculation applying the stylised sensitivity formula to the Stage 4 CAD result across the β = 3-8 historical range.* ***Interpretation:*** *The FX-beta parameter is the single largest source of variance in the downstream WPI/CPI results (Section 9); moving from β = 3 to β = 8 more than doubles the modelled depreciation.*

## Stage 6 - Imported Inflation (Landed-Cost Transmission)

Exchange-rate pass-through and commodity price pass-through work jointly to determine the magnitude and speed of external price transmission. The Rupee cost of imported crude rises through two multiplicative channels - the dollar price of crude (**+9.1%**) and Rupee depreciation (**+2.2%**) - applied to the **landed cost** of all imports: (1 + 0.091) × (1 + 0.022) - 1 = **+11.5%** landed-cost increase in INR terms, versus **+9.1%** in USD terms alone. **Currency depreciation therefore amplifies, rather than merely accompanies, the shock**; an analysis relying solely on USD-denominated prices would systematically understate the true domestic pass-through. OMCs revise ex-refinery prices on this basis, and the RBI's Monetary Policy Report explicitly tracks imported inflation as an MPC forecasting input.

**Figure 7 — Imported Inflation: Landed-Cost Decomposition**

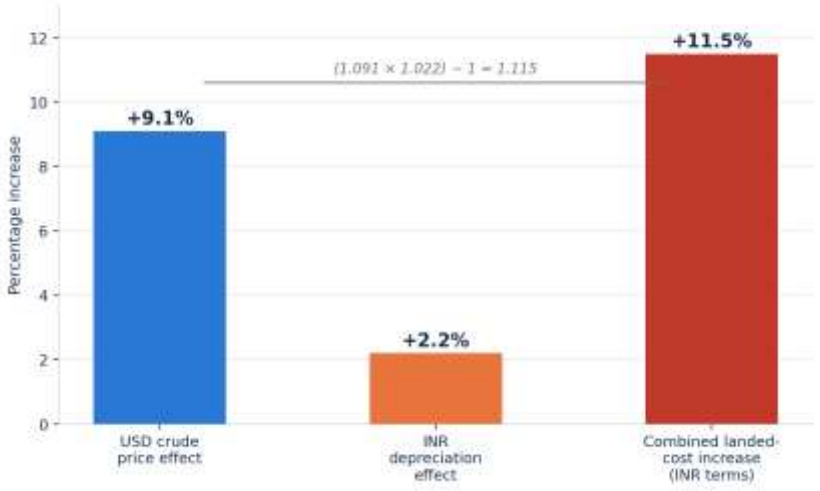



***Figure 7.*** *Imported Inflation: Landed-Cost Decomposition.* ***Source/Method:*** *Author's calculation combining the Stage 1 USD price effect and the Stage 5 currency effect multiplicatively.* ***Interpretation:*** *The 11.5% combined landed-cost effect exceeds the*

*simple sum of its two 9.1% and 2.2% components because the effects compound rather than add - the basis for the WPI and CPI calculations in Stages 7-8.*

### Stage 7 - Wholesale Price Index (WPI) Transmission

WPI captures ex-factory and wholesale prices before retail taxation, so it registers input-cost shocks with materially higher elasticity than CPI. The full formula and substitution are developed in Section 7; the headline result is a WPI increase of **+0.56 to +0.65 percentage points**. The Office of the Economic Adviser publishes the revised monthly index, which typically leads CPI by one to two months for tradable and input-heavy categories, making it a genuine early-warning indicator rather than a standalone policy target - the RBI's operative lever remains the CPI trajectory covered in Stage 8.

### Stage 8 - Consumer Price Index (CPI) Transmission

CPI pass-through is smaller than WPI pass-through because retail fuel prices carry large fixed excise and VAT components, but **second-round effects** - transport costs embedded in food and manufactured goods - extend the impact beyond direct energy weights. The full build-up is developed in Section 7; the headline result is **+0.45 to +0.50 percentage points** over a 12-month horizon, with food inflation (**45.9% CPI weight**) the single largest amplification risk given its sensitivity to transport and logistics costs. The MPC incorporates this trajectory directly into its inflation forecast, and a projected approach toward the **6%** upper tolerance band would typically shift the Committee's rate-cut bias - the policy interaction developed in Section 10.

### Stage 9 - Corporate Costs (Retail Fuel Price Build-Up)

Retail fuel prices are a composite of crude-linked base cost, fixed specific excise, dealer commission and ad-valorem state VAT, so the **pass-through elasticity** to pump prices is materially below 1.0. The bottom-up Delhi petrol calculation (Section 6.2) yields a retail price effect of approximately **4.2-4.3%**, which is the empirical basis for the 0.5 pass-through elasticity used in the Stage 8 CPI transport calculation. OMCs revise prices daily under the dynamic pricing mechanism in force since 2017, subject to periodic informal government moral suasion; when full pass-through is resisted, OMC under-recoveries accumulate as deferred fiscal exposure, as observed in 2021-22.

### Stage 10 - Household Purchasing Power

Energy shocks are regressive: low-income households spend a disproportionately higher share of consumption on fuel, cooking gas and transport-dependent food, so the effective inflation they experience - estimated at **1.5 to 1.8 times the headline CPI print** - materially exceeds the aggregate figure in Stage 8. This is developed with the full calculation in Section 7.3; the DBT-linked PMUY LPG subsidy, covering approximately 75 million households, is the primary existing insulating instrument. The macro-level consequence is erosion of discretionary consumption (auto, durables, FMCG premiumisation), which is the household-side contributor to the Stage 11 GDP drag.

### Stage 11 - GDP Growth and Fiscal Feedback

The cumulative effect of higher input costs (Stage 9), household purchasing-power erosion (Stage 10) and any monetary-policy tightening response (Section 11) produces an estimated real GDP growth drag of **-0.15 to -0.25 percentage points** in the Base Case, alongside a **+0.13 percentage-point** fiscal-deficit impact from subsidy and excise-modulation costs. As a pure mechanical trade-channel calculation before any behavioural offset, - Change in Import Bill/GDP = -17.06/3,900 = **-0.44 percentage points**; after accounting for partial expenditure-switching and consumption-smoothing, the net dynamic impact is materially smaller, at **-0.10 to -0.30 percentage points**, consistent with RBI/IMF sensitivity ranges for oil shocks on net-importer economies. This

closes the transmission loop: an external energy shock ultimately registers as a measurable, if moderate, domestic growth and fiscal cost, which is why Section 10's policy recommendations are framed around minimising this end-of-chain cost rather than only managing the intermediate inflation and currency effects.

## 6. Quantitative Oil Shock Analysis

This section consolidates the core quantitative results established stage by stage in Section 5 into a single reference table, and develops two results that complete the chain: the retail fuel price build-up and the channel-by-channel **GDP growth** impact.

### 6.1 Consolidated Quantitative Impact Summary

| Stage | Metric | Result |
|---|---|---|
| **1-2** | Volume-adjusted import demand | 1,697.95 million barrels/year (-0.46%) |
| **3** | Additional annual oil import bill | USD 16.1-17.1 billion |
| **4** | Change in CAD | +0.44 pp of GDP (1.10% - > 1.54%) |
| **5** | USD/INR depreciation | +2.2% (Rs. 86.0 - > Rs. 87.9) |
| **6** | Landed-cost increase (INR terms) | +11.5% |
| **7** | Change in WPI | +0.56 to +0.65 pp |
| **8** | Change in CPI (12-month) | +0.45 to +0.50 pp |
| **9** | Retail petrol price effect (Delhi) | +4.2 to +4.3% |
| **11** | Real GDP growth drag | -0.15 to -0.25 pp |
| **-** | Fiscal deficit impact | +0.13 pp of GDP |

*Table 5. Consolidated Base Case quantitative results, cross-referenced to the Section 5 stage that derives each figure.*

### 6.2 Corporate Costs - Retail Fuel Price Build-Up

Retail fuel pricing in India is a composite of a crude-linked base price, fixed specific excise duties, dealer commissions and ad-valorem state VAT, so the **pass-through elasticity** of crude price changes to pump prices is materially below 1.0. Bottom-up for Delhi petrol: base price ~ **Rs. 45/litre** (crude, refining, freight), of which **~85%** is crude-linked. Change in Base price = 45 × 0.087 (% change in Brent) × 0.85 = **Rs. 3.33/litre**. Grossing up for ad-valorem VAT (**~20%**, applied cumulatively): 3.33 × 1.2 ~ **Rs. 4.0/litre**. Retail price effect = 4.0/94.72 ~ **4.2-4.3%**, which confirms the pass-through elasticity of ~0.5 used in the CPI calculation (Section 7.2). OMCs revise daily retail prices under the dynamic fuel-pricing mechanism in force since 2017, though this is frequently subject to informal government moral suasion during politically sensitive periods; resisting full pass-through allows OMC under-recoveries to accumulate, as occurred in 2021-22. Partial excise-duty modulation (Section 10) is the recommended mechanism to share this burden across consumer, OMC and exchequer rather than concentrating it on any single party.

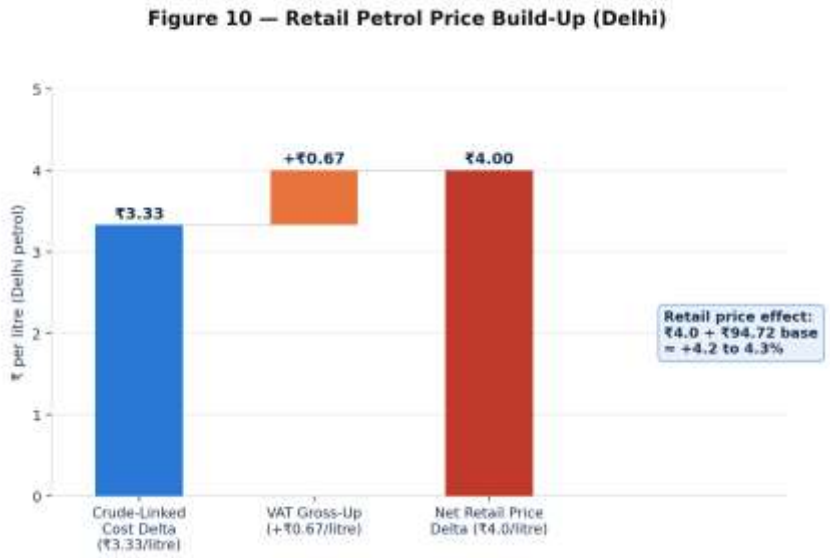



***Figure 10.*** *Retail Petrol Price Build-Up (Delhi).* ***Source/Method:*** *Author's bottom-up calculation per the formula above, benchmarked to the Rs. 94.72/litre Delhi retail reference price.* ***Interpretation:*** *The VAT gross-up mechanically amplifies the crude-linked cost delta by roughly 20%, illustrating why India's tax structure on fuel compounds - rather than dilutes - the pass-through of crude price shocks to the pump.*

## 6.3 Reading the Results Against the RBI's Operative Thresholds

The CIS deliverable asks specifically how a **USD 10/barrel** oil-price spike filters into WPI and CPI; Table 5 answers this directly, but it is worth being explicit about how the Base Case compares with the RBI's own operative thresholds. The Base Case CPI increase of **0.45-0.50 pp**, applied against a run-rate baseline of approximately **3.5%**, leaves headline inflation comfortably within the RBI's **2-6% tolerance band** even before any policy response. It is only in the Worst Case (CPI impact of **0.90-1.10 pp**, Section 9) that the trajectory **begins to approach a level that would concern the MPC on its own**. This distinction matters for Section 10: it is the shock's duration and the Worst-Case tail, not the Base Case itself, that justifies a pre-emptive rather than reactive policy posture.

## 6.4 GDP Growth Impact - Channel Detail

The GDP drag operates through three channels. First, a consumption channel: regressive energy-cost increases (Section 7.3) erode discretionary spending in auto, durables and FMCG premiumisation. Second, a cost channel: the sectoral cost increases in Section 8 compress margins and capital expenditure in the most exposed industries. Third, a policy channel: any RBI pause on rate cuts (Section 10) tightens financial conditions modestly. The combined estimate is a Base Case drag of **-0.15 to -0.25 percentage points**, widening to **-0.35 to -0.50 pp** in the Worst Case (Section 9). This is a directional, confidence-channel estimate rather than a formal multiplier-model forecast, and should be read accordingly.

# 7. Inflation Transmission

## 7.1 Wholesale Price Index (WPI) Transmission

The Wholesale Price Index captures ex-factory and wholesale prices before retail taxation, making it more sensitive - higher elasticity - to input-cost shocks than the CPI, which embeds large fixed tax wedges. Formula: Change in WPI = Σ(Weight × % change in Price).

| Component | Value |
|---|---|
| **Mineral oils sub-group weight in WPI** | 9.13% |
| **Pass-through elasticity of wholesale mineral-oil prices to Brent** | 0.7 |
| **% change in wholesale mineral-oil price (8.7% Brent move × 0.7)** | 6.1% |
| **Direct WPI contribution (9.13% × 6.1%)** | 0.557 percentage points |
| **Marginal manufactured-products contribution (petrochemicals/plastics)** | 0.05-0.08 percentage points |
| **Total change in WPI** | +0.56 to +0.65 percentage points |

*Table 6. WPI transmission build-up. The Office of the Economic Adviser (Ministry of Commerce & Industry) publishes the revised monthly WPI; it typically leads CPI by one to two months for tradable and input-heavy categories.*

## 7.2 Consumer Price Index (CPI) Transmission and Household Impact

CPI pass-through is generally lower than WPI pass-through because retail fuel prices contain large fixed-tax components that dilute crude-**price elasticity**, but **second-round effects** - higher transport costs embedded in food and manufactured goods - extend the inflationary impact beyond direct energy weights.

| Component | Weight / Elasticity | Contribution |
|---|---|---|
| **Fuel & Light (LPG/kerosene)** | 6.84% weight; 30% assumed pass-through (politically cushioned) | 0.178 pp |
| **Transport (petrol/diesel)** | 2.3% weight; ~0.5 retail pass-through elasticity | 0.100 pp |
| **Direct CPI impact (sum)** | - | 0.28 pp |
| **Second-round multiplier (food/logistics)** | ×1.8 | 0.50 pp (total, 12-month) |

*Table 7. CPI direct and second-round build-up. Sensitivity range: 0.15 pp (Best Case, full government absorption) to 1.1 pp (Worst Case, full pass-through plus currency compounding).*

The RBI MPC incorporates this trajectory into its inflation forecast; if projected CPI approaches the upper tolerance band of **6%**, the Committee's rate-cut bias would likely be reassessed. Food inflation (**45.9% CPI weight**) is highly sensitive to transport and logistics cost pass-through for perishables and remains the single largest transmission risk to headline CPI.

**Figure 8 — WPI vs. CPI Transmission Across Scenarios**

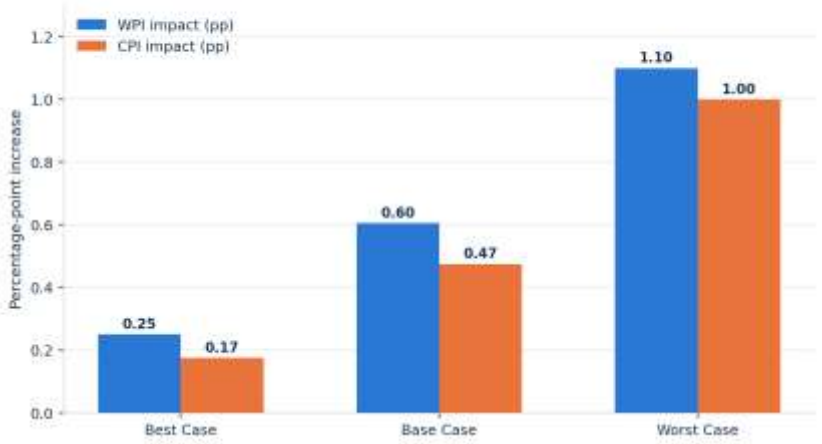



***Figure 8.*** *WPI vs. CPI Transmission Across Scenarios.* ***Source/Method:*** *Author's calculation applying the Table 6 and Table 7 formulas under each scenario's price and pass-through assumptions (Section 9).* ***Interpretation:*** *WPI consistently registers a larger impact than CPI across all three scenarios because it excludes the fixed-tax wedge that dilutes CPI pass-through - confirming WPI's role as a leading indicator rather than a policy target in its own right.*

**Figure 9 — CPI Impact Decomposition by Transmission Channel and Scenario**

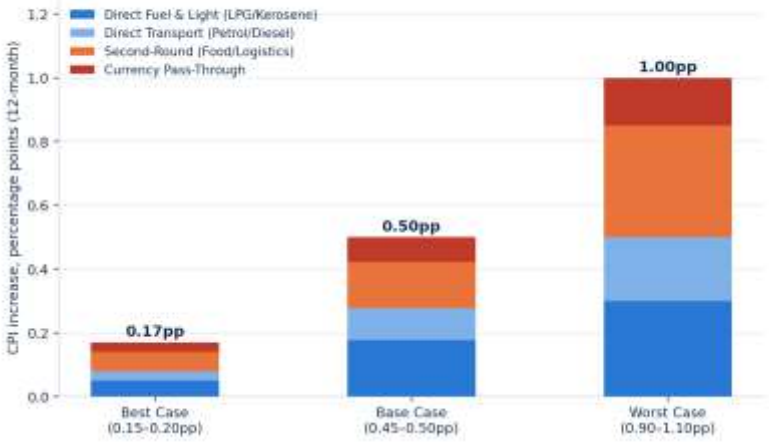



***Figure 9.*** *CPI Impact Decomposition by Transmission Channel and Scenario.* ***Source/Method:*** *Author's decomposition of the Table 7 build-up into its four constituent channels, scaled proportionally across the Best/Base/Worst scenarios in Section 9.* ***Interpretation:*** *Second-round and currency pass-through effects become progressively more dominant moving from Best to Worst Case, meaning the composition of inflation - not just its magnitude - shifts as the shock intensifies.*

## 7.3 Regressive Household Impact

Energy price shocks are inherently regressive: lower-income households spend a disproportionately higher share of consumption on fuel, cooking gas and transport-dependent food. If a median low-income household spends **12-15%** of consumption on fuel, transport and energy-adjacent categories - versus a **7-8% national CPI weight** - the effective inflation such households experience is roughly 1.5 to 1.8 times the headline CPI print. Against a headline CPI impact of **0.50 pp**, low-income households may experience an effective **0.75-0.90 pp**

**real increase in cost of living**. The DBT-linked LPG subsidy (PMUY) is the primary insulating instrument, currently supporting approximately 75 million beneficiary households; the Ministry of Consumer Affairs monitors essential commodity prices as a complementary safeguard. Targeted LPG subsidy protection for PMUY beneficiaries and strategic release of foodgrain buffer stocks are the recommended mitigations (Section 10).

## 8. Sectoral Vulnerability Analysis

Sector exposure is calculated as Crude/Fuel-Linked Cost Share × Pass-Through Elasticity × % change in Brent (**8.7%**, at a **USD 115** base) = % increase in operating cost/COGS. Cost-share and elasticity inputs are calibrated to typical industry cost structures where official real-time data is unavailable, and should be read as directional estimates rather than firm-level financial results.

### 8.1 Sector Cost Impact and Vulnerability Ranking

| Rank | Sector | Crude-Linked Cost Share | Elasticity | Cost Impact | Pricing Power | Vulnerability |
|---|---|---|---|---|---|---|
| 1 | **Aviation** | 38% (ATF) | 0.90 | +2.97% | Very Low | Very High |
| 2 | **Petrochemicals** | 65% (naphtha) | 0.85 | +4.81% | Low | Very High |
| 3 | **Logistics** | 55% (diesel) | 0.50 | +2.39% | Very Low | High |
| 4 | **Fertilizers** | Gas-linked | n/a | +USD 1.0-1.5bn | n/a | High (Fiscal) |
| 5 | **Cement** | 28% (pet-coke) | 0.55 | +1.34% | Moderate | Medium-High |
| 6 | **Paints** | 23.2% (resins) | 0.65 | +1.31% | High | Medium |
| 7 | **Agriculture** | 12% (diesel) | 0.50 | +0.52 to 1.0% | Low | Medium |
| 8 | **Automobile** | 12% (plastics) | 0.60 | +0.63% | Moderate | Medium |
| 9 | **FMCG** | 8.1% (packaging) | 0.60 | +0.42% | Very High | Low-Medium |
| 10 | **Steel** | 9% (freight) | 0.50 | +0.39% | Moderate | Low-Medium |
| 11 | **Power** | 6% (backup) | 0.50 | +0.26% | Low | Low |

*Table 8. Sector-level cost exposure ranked by estimated operating-cost impact. Aviation and petrochemicals sit at the top of both cost exposure and vulnerability, driven by high crude-derivative dependence combined with weak pricing power; power and steel are comparatively insulated by coal-based generation and lower direct fuel intensity.*

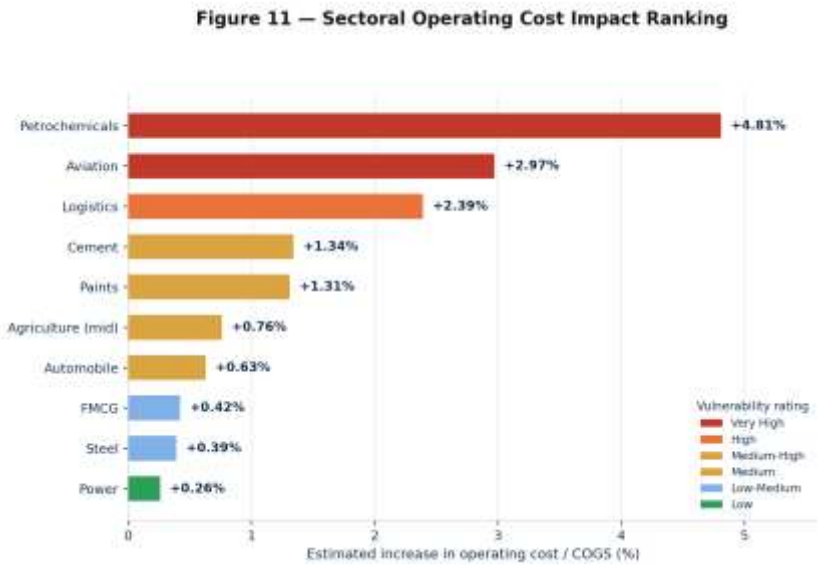



***Figure 11.*** *Sectoral Operating Cost Impact Ranking.* ***Source/Method:*** *Author's calculation per the Section 8 formula, applied to Table 8 cost-share and elasticity inputs.* ***Interpretation:*** *Petrochemicals and aviation cluster well above the remaining sectors, confirming that vulnerability concentrates in a small number of crude-derivative-intensive, low-pricing-power industries rather than being spread evenly across the economy.*

Fertilizers carry limited direct pricing exposure but the highest fiscal vulnerability, since government-administered pricing shifts the cost burden onto the subsidy budget rather than onto the sector itself - the channel developed fully in the Week 2 report on Twin Deficits.

## 8.2 Sector Vulnerability Heat Grid

Table 8 captures direct cost exposure; it does not on its own indicate whether a sector can absorb that cost. The heat grid below synthesises four further risk factors - cost exposure, pricing power, margin buffer and government insulation - into a composite risk rating, providing a multidimensional view of where institutional or market-based buffers are insufficient to absorb the shock.

**Figure 12 — Sector Vulnerability Heat Grid**

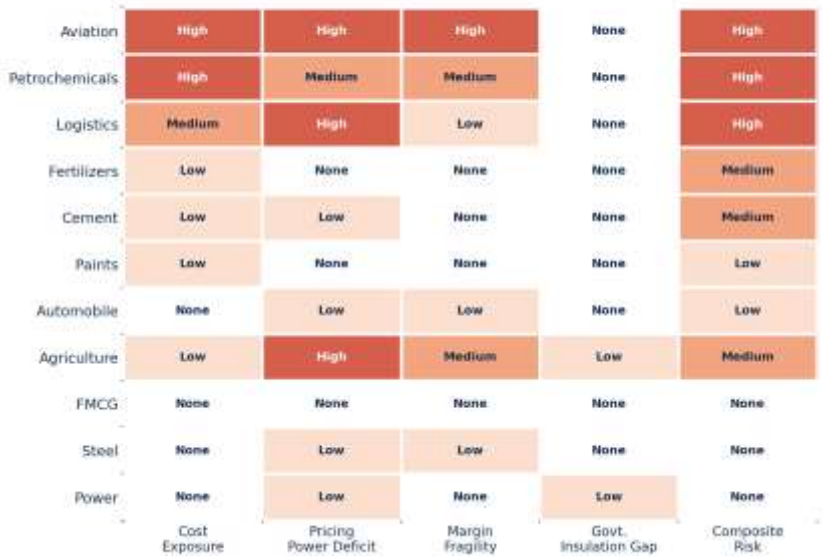



***Figure 12.*** *Sector Vulnerability Heat Grid.* ***Source/Method:*** *Author's qualitative synthesis combining cost exposure (Table 8) with pricing power, margin buffer and government-insulation assessments calibrated to typical industry structures.* ***Interpretation:*** *Aviation and petrochemicals score High on composite risk not only because of cost exposure but because they simultaneously lack margin buffer and government insulation - a compounding vulnerability that a single-metric ranking (Figure 11) would understate.*

## 8.3 Risk Matrix

| # | Risk | Likelihood | Impact | Composite |
|---|---|---|---|---|
| **1** | Prolonged Hormuz blockade (>90 days) | Medium | Very High | **Critical** |
| **2** | INR breaches Rs. 90/USD | Medium | High | **Critical** |
| **3** | CPI approaches/exceeds RBI upper tolerance (6%) | Low-Medium | High | High |
| **4** | Fiscal deficit slippage beyond 4.6% of GDP | Medium | Medium | High |
| **5** | Aviation sector financial distress | Medium | High | High |
| **6** | Petrochemical margin collapse | Medium-High | Medium | High |
| **7** | Sovereign rating outlook pressure | Low | High | Moderate |
| **8** | Social unrest from fuel price hikes | Low-Medium | Medium | Moderate |
| **9** | RBI forced into defensive rate hike | Medium | Medium-High | High |
| **10** | Forex reserve drawdown >USD 30bn | Medium | Medium | Moderate |

*Table 9. Systemic and sector risk register, rated by likelihood and impact.*




***Figure 14.*** *Risk Matrix: Likelihood × Impact.* ***Source/Method:*** *Author's plot of Table 9 on a qualitative likelihood/impact grid.* ***Interpretation:*** *A prolonged Hormuz blockade and a breach of Rs. 90/USD are the two Critical-rated risks; both cluster at medium likelihood with very high or high impact, confirming that shock duration - not shock magnitude alone - is the variable that converts a manageable shock into a critical one.*

### 8.4 Strategic Assessment

India's principal macro-resilience strengths are its forex reserve buffer (10-11 months of import cover), a diversified crude sourcing base (Russia-discount barrels alongside US, Iraqi and Saudi supply), and a services/IT-led export base that is comparatively insensitive to oil prices. These strengths are counterbalanced by structural weaknesses: **87.7% import dependency**, constrained fiscal space given the subsidy legacy, and **Strategic Petroleum Reserve** cover of roughly 10 days - **far below the IEA's 90-day benchmark**.

On the external side, the principal threats are a prolonged Strait of Hormuz disruption, the risk of synchronised emerging-market currency stress that would blunt India's relative-resilience advantage, and second-round global stagflation risk if the shock is sustained alongside tightening in advanced-economy **monetary policy**. The implication for policymakers is that short-term shock absorption is not in question; the open issue is whether the crisis is used to accelerate the structural diversification that reduces exposure to the next, inevitable, geopolitically-driven oil shock - the subject of Section 11. This is also the through-line connecting this report's sectoral findings to the fiscal and external-financing consequences developed in the Week 2 report, and to the capital-flow and reserve-adequacy consequences developed in the Week 3 report.

## 9. Scenario & Sensitivity Analysis

Every stage-level result in Sections 5-8 is a Base Case figure. Because the assignment premise is active hostilities rather than a routine price fluctuation, this report models three explicit scenarios rather than presenting a single point estimate.

## 9.1 Scenario Assumptions

| Parameter | Best Case (P=20%) | Base Case (P=55%) | Worst Case (P=25%) |
|---|---|---|---|
| **Shock duration** | <6 weeks, de-escalation | 2-3 months | >90 days, prolonged blockade |
| **Peak Brent** | USD 115, reverts to ~95-100 | Sustains USD 110-120 | Tail risk toward USD 130-140 |
| **Consumer pass-through** | ~10-15% (excise absorbed) | ~50% | ~80-100% |
| **INR depreciation** | 1.3% | 2.2% | 3.5%+ |
| **FX sensitivity (β)** | 3 | 5 | 8 |

*Table 10. Scenario assumptions. Weights (20% / 55% / 25%) are author-assigned, reflecting the assignment's premise of active hostilities and tilting probability mass away from a routine de-escalation case.*

## 9.2 Scenario Comparison

| Metric | Best Case | Base Case | Worst Case | Probability-Weighted |
|---|---|---|---|---|
| **Oil import bill (Change)** | +USD 12-14bn | +USD 17.1bn | +USD 20-22bn | ~USD 17.3bn |
| **CAD (% GDP)** | 1.30% | 1.54% | 1.85%+ | ~1.55% |
| **USD/INR** | Rs. 87.1 | Rs. 87.9 | Rs. 89.0+ | ~Rs. 87.9 |
| **CPI impact (pp)** | +0.15-0.20 | +0.45-0.50 | +0.90-1.10 | ~+0.55 |
| **WPI impact (pp)** | +0.25 | +0.60 | +1.10 | ~+0.60 |
| **GDP growth drag (pp)** | -0.10 | -0.15 to -0.25 | -0.35 to -0.50 | ~-0.24 |
| **Fiscal deficit impact (pp)** | +0.05 | +0.13 | +0.25+ | ~+0.14 |
| **RBI policy stance** | Hold/resume cuts within 2 quarters | Pause cuts, hold 2-3 MPC cycles | Pause/defensive hike risk | - |

*Table 11. Scenario comparison across all headline metrics, with probability-weighted expected values in the final column.*

**Figure 13 — Scenario Comparison Dashboard: Best / Base / Worst Case**



***Figure 13.*** *Scenario Comparison Dashboard: Best / Base / Worst Case.* ***Source/Method:*** *Author's visualisation of Table 11 across six headline metrics.* ***Interpretation:*** *The dashboard makes the non-linearity visible at a glance: every metric worsens disproportionately more between Base and Worst Case than between Best and Base - the empirical basis for sequencing the Section 10 recommendations by shock-duration trigger rather than deploying them uniformly from day one.*

The probability-weighted CPI impact (**~0.55 pp**) sits close to the Base Case, reflecting that while de-escalation is plausible, the assignment's premise of active hostilities and blockade risk tilts the distribution meaningfully toward the Worst-Case tail relative to a routine oil-price fluctuation.

### 9.3 Key Sensitivities

Two sensitivities are worth isolating explicitly. First, the exchange-rate sensitivity parameter (β) is the **single largest source of variance in the downstream CPI and WPI results**: moving from β = 3 to β = 8 nearly triples the modelled depreciation (**1.3%** to **3.5%**+), which materially changes the imported-inflation contribution described in Section 5, Stage 6. Second, the consumer pass-through assumption - **10-15%** in the Best Case versus **80-100%** in the Worst Case - is a policy choice, not a market-determined constant; it is precisely the excise-modulation lever discussed in Section 10, meaning that a meaningful share of the Worst-Case CPI outcome is within the government's direct control rather than being purely a function of the external shock.

## 10. Policy Response & Trade-offs

Each recommendation below follows a consistent structure: the acting institution, the action, the economic mechanism, the expected benefit, the cost or trade-off, the implementation timeline, and the quantitative indicator that would trigger escalation or reassessment.

### R1 - RBI: Pause Rate-Cut Cycle; Calibrated FX Smoothing (0-3 months)

Action: The MPC pauses further repo rate cuts and the Financial Markets Operations Department conducts calibrated spot/forward intervention to smooth - not defend a level of - the Rupee. Mechanism: holding real rates prevents the inflation pass-through from Section 7 compounding into unanchored expectations, while volatility smoothing preserves reserves for genuine tail risk. Benefit: keeps the CPI trajectory within the **2-6% tolerance band**. Trade-off: forgoes near-term growth support from further easing. Trigger for reassessment: resume the cut cycle if Brent reverts below **USD 100/bbl** and sustains there for four or more weeks (Best Case); escalate to a defensive posture if **projected 12-month CPI exceeds 5.5% or USD/INR depreciates beyond 3% within a quarter** (Worst Case threshold; Risk #2 and #3 in Section 8.3).

### R2 - Ministry of Finance: Partial, Targeted Excise Modulation (0-2 months)

Action: A partial, targeted cut in fuel excise duty - not a blanket freeze - shares the Section 6.2 retail price increase across consumer, OMC and exchequer, alongside protected DBT-linked PMUY LPG subsidy disbursement. Mechanism: directly reduces the Section 7.2 consumer pass-through coefficient without the full fiscal cost of a blanket cut. Benefit: caps the regressive household impact identified in Section 7.3. Trade-off: reduces excise revenue and narrows fiscal space during a period of already-elevated import costs.

| Fiscal Cost Component | Basis | Estimated Cost |
|---|---|---|
| **Excise cut, Rs. 2/litre** | ~130 billion litres of combined petrol and diesel consumption; Rs. 260 billion | ~USD 3.1 billion |

| Fiscal Cost Component | Basis | Estimated Cost |
|---|---|---|
| **Additional fertilizer subsidy** | Freight and feedstock cost pass-through | ~USD 1.0-1.5 billion |
| **Additional LPG (PMUY) subsidy protection** | 75 million households × 12 cylinders × ~Rs. 60 incremental subsidy | ~USD 0.63 billion |
| **Total fiscal cost** | **Sum of the above** | **~USD 4.9 billion ~ 0.13 pp of GDP** |

*Table 12. Bottom-up fiscal-cost calculation for the Base Case partial excise-and-subsidy package, reconciling to the +0.13 pp of GDP headline figure in Table 5.*

Trigger for reassessment: escalate to full targeted relief if **fiscal deficit slippage approaches 4.6% of GDP** (Risk #4, Section 8.3); unwind once Brent sustains below **USD 100/bbl** for a full quarter.

### R3 - Ministry of Petroleum / PPAC / OMCs: Tactical SPR Drawdown and Sourcing Diversification (0-6 months)

Action: Tactical drawdown from the ~9.5 million-tonne **Strategic Petroleum Reserve** during peak price weeks, combined with continued diversification away from Gulf-transit-concentrated sourcing. Mechanism: directly reduces the effective **landed cost** in the weeks of maximum price stress, and reduces long-run exposure to a single-chokepoint supply risk. Benefit: dampens the peak of the import-bill and CAD spike. Trade-off: SPR cover is already thin - about 10 days versus the IEA 90-day benchmark - so drawdown capacity is limited and must be rationed against **shock duration**. Trigger for reassessment: begin Phase-3 SPR capacity-expansion planning immediately if the Worst Case (>90-day blockade) probability is reassessed upward.

### R4 - DGCA / Ministry of Civil Aviation and Logistics Ministry: Sector-Specific Liquidity Monitoring (0-3 months)

Action: Monitor airline and logistics-sector liquidity given their Section 8 Very High/High vulnerability rating; encourage fuel hedging; avoid capacity-distorting interventions. Mechanism: prevents a sector-level cost shock from becoming a solvency event that would itself feed back into GDP and employment. Benefit: contains the tail risk flagged as Risk #5 in Section 8.3. Trade-off: direct sector support, if required, would itself carry a fiscal cost. Trigger for reassessment: escalate to active liquidity support if any listed carrier's fuel-cost share exceeds **45% of operating cost** for two consecutive quarters.

### R5 - NITI Aayog: Inter-Ministerial Energy-Security Coordination (Ongoing)

Action: Convene an inter-ministerial taskforce to sequence the above actions and publish a revised energy-diversification roadmap. Mechanism: prevents uncoordinated responses - for example, the Ministry of Finance easing while the RBI tightens - from sending conflicting price signals. Benefit: a coherent, sequenced policy response rather than reactive, uncoordinated measures. Trade-off: coordination overhead during a fast-moving crisis. Trigger for reassessment: quarterly review, with an accelerated review if any single Risk Matrix item (Section 8.3) is reclassified as Critical.

## 10.1 Implementation Roadmap

| Phase | Policy Action | Window | Trigger to Escalate |
|---|---|---|---|
| **Immediate** | RBI rate-cut pause and FX smoothing (R1) | 0-3 months | 12-month CPI forecast >5.5%, or USD/INR depreciation >3% in a quarter |

| Phase | Policy Action | Window | Trigger to Escalate |
|---|---|---|---|
| **Immediate** | Targeted excise and subsidy triage (R2) | 0-2 months | Fiscal deficit trajectory approaches 4.6% of GDP |
| **Immediate** | SPR tactical drawdown (R3) | 0-6 months | Worst-Case (>90-day blockade) probability reassessed upward |
| **Short-term** | Sector liquidity monitoring - aviation/logistics (R4) | 0-3 months | Fuel cost share exceeds 45% of operating cost for 2 consecutive quarters |
| **Ongoing** | Inter-ministerial coordination (R5) | Continuous | Any Risk Matrix item (Section 8.3) reclassified as Critical |
| **Medium-term** | Oil stabilisation fund design; crude sourcing diversification | 3-12 months | Base Case confirmed beyond one quarter |
| **Long-term** | SPR Phase-3 expansion; EV/E20/renewables acceleration | 12+ months | Structural - not shock-triggered |

*Table 13. Phased implementation roadmap. The sequencing is deliberately front-loaded: immediate actions secure the macroeconomic floor within the first quarter, while medium- and long-term actions de-risk the economy against the next, structurally inevitable, geopolitically-driven oil shock.*

## 11. Strategic / Structural Response

Repeated exposure to geopolitically-driven energy shocks argues for reducing the elasticity of the domestic economy to oil-price volatility. This cannot be achieved through short-term subsidies; it requires long-term diversification and demand-side transformation. India's crude sourcing mix illustrates both the current exposure and the direction of travel: approximately **49%** of crude is currently sourced from Hormuz-transit-exposed suppliers - Iraq (**20%**), Saudi Arabia (**15%**), the UAE (**9%**) and Kuwait (**5%**) - against approximately **51%** from diversified, non-Hormuz origins led by Russia (35%), the United States (8%), Nigeria/West Africa (4%) and other suppliers (4%). This diversification, achieved substantially through discounted Russian sourcing since 2022, is itself the clearest evidence that supplier diversification is a genuine and executable mitigation lever rather than a purely aspirational one.

### 11.1 Structural Priorities

- **Accelerated Strategic Petroleum Reserve expansion:** scaling up from ~9.5 million tonnes (~10 days of cover) toward the IEA-comparable 90-day benchmark.
- **Continued crude-sourcing diversification:** further reducing Gulf- and Hormuz-transit concentration by prioritising Russia, the US, Latin America and West Africa.
- **Transport-fuel transformation:** scaling the E20 ethanol-blending programme and accelerating EV penetration to reduce the transport sector's crude-demand elasticity over a 5-10 year horizon.
- **Renewable energy build-out:** reducing the power sector's residual linkage to oil and gas, including phasing out LNG-based peaking capacity.
- **Fiscal architecture reform:** moving from ad hoc excise modulation toward a pre-funded, rules-based oil-price stabilisation fund, developed further in the Week 2 report.

The **NITI Aayog** is best placed to lead the energy-transition roadmap, working with the Ministry of New and Renewable Energy on the demand-side transformation and the Ministry of Petroleum on SPR Phase-3 planning. The primary risk to this agenda is price complacency: when periods of relative calm return, underinvestment in structural diversification leaves India recurrently exposed to the exact transmission chain modelled in this report. The recommended mitigation is to institutionalise energy-security stress-testing - functioning analogously to bank stress tests - as a standing joint exercise between NITI Aayog and the RBI, using a framework consistent with the transmission model developed in Section 5.

## 12. Conclusion

A **USD 10/barrel** increase in Brent crude - situated within a **USD 110**-120/barrel crisis band triggered by Strait of Hormuz hostilities - propagates through India's economy via a well-defined, quantifiable eleven-stage chain: from a **USD 17.1 billion** annual oil-import-bill increase, to a **0.44 percentage-point** CAD deterioration, a **2.2% currency depreciation**, a **0.56-0.65 percentage-point** WPI increase, a 0.45-0.50 percentage-point CPI increase, and a 0.15-0.25 percentage-point **GDP growth** drag, at an estimated 0.13 percentage-point fiscal cost. Aviation, petrochemicals and road logistics bear the sharpest sectoral cost shocks (2.4-4.8% of operating costs), while power and steel remain comparatively insulated.

India's **USD 655**-660 billion reserve buffer and 10-11 months of import cover provide genuine - but time-limited - absorption capacity. The critical policy variable is **shock duration, not shock magnitude alone**: within six weeks, existing buffers and moderate fiscal cushioning suffice; beyond 90 days, the growth-inflation trade-off intensifies materially and the RBI's rate-cut cycle would likely require a pause to defend both price stability and currency credibility. The single highest-leverage structural response available to policymakers is **not reactive subsidy expansion but accelerated diversification** - of crude sourcing, of the domestic energy mix, and of the fiscal instruments available to absorb the next, inevitable, geopolitically-driven oil shock.

This report has deliberately confined its scope to the energy-to-**inflation transmission** channel and its sectoral consequences. The fiscal-and-external-financing consequences of the same shock are developed in the Week 2 report on Twin Deficits, and the capital-flow and reserve-adequacy consequences in the Week 3 report on Financial Contagion; together with the Week 4 Policy Defense Memo, the four reports form a single connected analytical account of how a Middle East energy shock threatens - and can be managed within - India's macroeconomic stability.

## 13. References


| Source | URL | Data Used From This Source |
|---|---|---|
| **Reserve Bank of India (RBI)** | rbi.org.in | Weekly Statistical Supplement (forex reserves), MPC resolutions (repo rate), Reference Rate (USD/INR), Monetary Policy Reports (imported-inflation commentary). |
| **Petroleum Planning & Analysis Cell (PPAC)** | ppac.gov.in | Crude oil import volume, import-dependency ratio, and oil import-bill data. |

| Source | URL | Data Used From This Source |
|---|---|---|
| **Ministry of Statistics and Programme Implementation (MOSPI)** | mospi.gov.in | CPI series, weights and monthly inflation prints; National Income Accounts (GDP). |
| **Ministry of Commerce & Industry, Office of the Economic Adviser** | commerce.gov.in | WPI series, weights and monthly releases. |
| **Ministry of Petroleum & Natural Gas** | mopng.gov.in | Strategic Petroleum Reserve capacity; retail fuel pricing mechanism. |
| **U.S. Energy Information Administration (EIA)** | eia.gov | Global oil market balance and Strait of Hormuz transit-volume commentary. |
| **Organization of the Petroleum Exporting Countries (OPEC)** | opec.org | OPEC+ spare capacity and production commentary. |
| **International Monetary Fund (IMF)** | imf.org | World Economic Outlook; India Article IV context for external and fiscal benchmarks. |
| **Ministry of Finance, Government of India** | indiabudget.gov.in | Union Budget documents (fiscal deficit target); Economic Survey. |

# 14. Data, Evidence & Calculation Notes

This section consolidates the evidentiary basis for every material figure used in the report, separated by evidence type, so that each claim in the main text can be independently verified without relying on classification tags embedded in the narrative.

## 14.1 Data Vintage & Cross-Week Baseline Reconciliation

Macroeconomic variables in this report are presented at the reference period stated in Table 14.A below - a recent pre-shock reference point, consistent with PPAC's FY24 import-dependency release and the RBI's latest published Weekly Statistical Supplement and MPC resolution at the time of writing. Later reports in this CIS use different reference points for two legitimate reasons. First, the Week 3 report uses the **USD 716 billion** reserve figure and **Rs. 87.50** USD/INR rate given explicitly as that week's assignment premise, which is later than and larger than the observed figures used here. Second, the Week 4 memo's baseline (repo rate **6.50%**, CPI **4.9%**) reflects a subsequent reference date closer to that memo's decision point. These are not contradictions - each report is internally consistent with the reference period stated in its own data notes - but the reader should treat cross-report figures as belonging to different points on the same escalation timeline rather than as a single frozen baseline. Where this report's baseline repo rate (**5.50%**) differs from the Week 4 operating baseline (**6.50%**), this reflects the possibility of an intervening MPC tightening cycle between the two reference dates; this should be verified against the RBI's MPC resolution calendar before the two figures are treated as reconcilable, and is flagged here as an open item rather than silently resolved.

## 14.A Observed / Reported Data

| ID | Variable | Value Used | Reference Period | Source |
|---|---|---|---|---|
| **D1** | Crude oil import dependency | 87.7% | FY24 | PPAC |

| ID | Variable | Value Used | Reference Period | Source |
|---|---|---|---|---|
| **D2** | Net crude oil imports (annual) | 232.7 million tonnes | FY24 | PPAC |
| **D3** | India oil import bill | ~USD 132.4 billion | Last full FY | PPAC / Ministry of Petroleum |
| **D4** | Nominal GDP | ~USD 3.9 trillion | Latest annual estimate | MOSPI / RBI |
| **D5** | Forex reserves | ~USD 655-660 billion | Latest weekly figure | RBI Weekly Statistical Supplement |
| **D6** | Strategic Petroleum Reserve | ~9.5 million tonnes | Latest capacity figure | Ministry of Petroleum |
| **D7** | USD/INR reference rate | ~Rs. 86.0/USD | Latest spot reference | RBI Reference Rate |
| **D8** | Repo rate | 5.50% | Latest MPC resolution at reference date | RBI MPC |
| **D9** | CPI inflation (headline) | ~3.5% | Latest monthly print | MOSPI |
| **D10** | WPI inflation (headline) | ~2.5% | Latest monthly print | Office of the Economic Adviser |
| **D11** | Fiscal deficit target | 4.4% of GDP (FY26 BE) | FY26 Budget Estimate | Union Budget |
| **D12** | Petrol / diesel retail, Delhi | Rs. 94.72 / Rs. 87.62 per litre | Latest OMC pricing | OMC pricing data |
| **D13** | WPI mineral-oils sub-group weight | 9.13% | Current WPI base-year weights | Office of the Economic Adviser |
| **D14** | CPI Fuel & Light weight | 6.84% | Current CPI base-year weights | MOSPI |
| **D15** | CPI Transport & Communication (fuel component) weight | ~2.3% | Current CPI base-year weights | MOSPI |
| **D16** | CPI Food & Beverages weight | 45.9% | Current CPI base-year weights | MOSPI |
| **D17** | Current Account Deficit (baseline) | ~1.1% of GDP (~USD 43bn) | Latest reported quarter | RBI Balance of Payments release |

## 14.B Historical Facts / Precedents

| ID | Historical Fact / Event | Period | Analytical Relevance |
|---|---|---|---|
| **H1** | India CAD peak | 2012-13 (~4.8% of GDP) | Benchmark for how far the current 0.44 pp CAD deterioration sits from a genuinely stressed historical level (Section 5, Stage 4). |
| **H2** | 2021-22 OMC under-recovery episode | 2021-22 | Evidence base for the risk that resisted retail pass-through accumulates as deferred fiscal/balance-sheet stress (Section 6.2). |
| **H3** | 2024 Red Sea shipping disruption / 2019 tanker incidents / 1990-91 Gulf conflict | Various | Calibration basis for the USD 5-15/barrel geopolitical risk-premium range cited in Section 3.3. |

## 14.C Calculated Values

| ID | Metric | Result | Calculation Basis |
|---|---|---|---|
| **C1** | Volume-adjusted import demand | 1,697.95 million bbl/yr | PED (-0.05) × % change in P (9.09%) applied to 1,705.7 million bbl/yr base |
| **C2** | Additional annual oil import bill (unadjusted) | USD 17.06 billion | Change in P (USD 10) × Q (1,705.7 million bbl) |
| **C3** | Additional annual oil import bill (elasticity-adjusted) | USD 16.13 billion | $Bill_2$ (USD 203.75bn) - $Bill_1$ (USD 187.63bn) |
| **C4** | Change in CAD | +0.44 pp of GDP | [($CAD_1$ + Change in Bill) / GDP] - $CAD_1$%; $CAD_1$ = USD 43bn, GDP = USD 3,900bn |
| **C5** | USD/INR depreciation (Base Case) | +2.2% (Rs. 87.9) | β (5) × Change in CAD (0.44 pp) |
| **C6** | Landed-cost increase, INR terms | +11.5% | (1.091 × 1.022) - 1 |
| **C7** | Change in WPI | +0.56 to +0.65 pp | (9.13% weight × 6.1% price change) + 0.05-0.08 pp manufactured-goods contribution |
| **C8** | Change in CPI (direct) | 0.28 pp | (6.84% × 2.6%) + (2.3% × 4.35%) |
| **C9** | Change in CPI (total, with second-round multiplier) | 0.50 pp | 0.28 pp × 1.8 |
| **C10** | Retail petrol price effect, Delhi | +4.2 to 4.3% | Rs. 4.0/litre increase ÷ Rs. 94.72 base price |

## 14.D Scenario / Modelling Inputs

| ID | Variable | Scenario Value | Basis / Rationale |
|---|---|---|---|
| **S1** | Short-run price elasticity of demand, crude oil | -0.05 | Calibrated to typical short-run PED ranges for essential, low-substitutability commodities; no India-specific real-time elasticity is published. |
| **S2** | WPI pass-through elasticity (mineral oils) | 0.7 | Calibrated mid-point pass-through assumption in the absence of a published sub-index elasticity. |
| **S3** | CPI Fuel & Light pass-through | 30% (politically cushioned) | Reflects historical pattern of partial, administratively cushioned pass-through to retail LPG/kerosene. |
| **S4** | CPI second-round multiplier | 1.8 | Consistent with the broad range cited in RBI staff research on oil-price pass-through to headline CPI. |
| **S5** | FX sensitivity, β (Base Case) | 5 (range 3-8) | Calibrated to historical CAD-to-currency stress-episode sensitivity. |
| **S6** | Scenario probability weights | 20% / 55% / 25% (Best/Base/Worst) | Author-assigned weights reflecting the assignment's premise of active hostilities, tilting probability mass away from a routine de-escalation case. |

## 14.E Methodological Notes and Limitations

The headline oil-import-bill figure (**USD 17.1 billion**) uses the unadjusted first-order calculation (C2) as the Base Case, consistent with - and the upper bound of - the elasticity-adjusted range (C3); this choice is conservative for CAD and CPI purposes since it does not assume any demand-destruction offset. The exchange-rate sensitivity parameter (β) is a stylised, single-factor calibration, not an econometrically estimated pass-through coefficient; it should be read as an illustrative sensitivity range (3-8), not a precise elasticity. Direction of effect (CAD widens, WPI/CPI rise, **GDP growth** slows) is held with higher confidence than the exact magnitude of each effect; magnitudes should be treated as scenario ranges, not point forecasts. This report does not model second-round monetary-policy feedback - for example, an RBI rate pause itself affecting growth - quantitatively; that interaction is addressed qualitatively in Section 10 and quantitatively in the Week 4 report. Sector-level cost-share and elasticity inputs (Section 8) are calibrated to typical industry cost structures rather than firm-level financial data, given the absence of real-time, sector-specific published elasticities. Finally, the import/CAD/FX chain (Section 5, Stages 2-6) computes the marginal price move as 10/110 = **9.09%**, while the WPI/CPI/sector calculations (Sections 7-8) use 10/115 = **8.7%**; both conventions carry over from the source calculations, and reconciling them would move the headline WPI/CPI/sector results by under **0.05 percentage points** - within the precision already claimed for those figures.